

Tulip Bouquets (tulips)

Anna really likes tulips. She has N tulips in her garden, numbered from 0 to $N - 1$. The *beautiness* of tulip i is T_i . She wants to create K (non-empty) bouquets from the tulips. To do that, she starts to walk from the first tulip towards the last. At each flower, she can either

- insert it into the current bouquet, or
- finish the current bouquet and start a new one. The current tulip is added to the new bouquet.

Note that, after finishing a bouquet she won't be able to insert more tulips into it!



Figure 1: Tulips are indeed beautiful.

The *beautiness* of a bouquet is the minimum of the *beautinesses* of the tulips in it. She wants to maximize the sum of the *beautinesses* of the K bouquets by partitioning the tulips optimally. Your task is to calculate this maximum value!

 Among the attachments of this task you may find a template file `tulips.*` with a sample incomplete implementation.

Input

The input file consists of:

- a line containing integers N , K .
- a line containing the N integers $T_0, \dots, T_{N-1}$.

Output

The output file must contain a single line with an integer M , the maximum total *beautiness* of the bouquets.

Constraints

- $1 \leq K \leq N \leq 100\,000$.
- $1 \leq N \cdot K \leq 50\,000\,000$.
- $0 \leq T_i \leq 1\,000\,000\,000$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (12 points) $K \leq 2$.

- **Subtask 3** (25 points) $N \leq 16$.

- **Subtask 4** (25 points) $N \leq 500$.

- **Subtask 5** (38 points) No additional limitations.


Examples

input	output
5 2 3 4 1 5 2	4
6 4 4 2 6 1 3 5	14

Explanation

In the **first sample case** $3\,4\,|\,1\,5\,2$ is an optimal way to distribute the flowers into 2 bouquets. The total *beautiness* is $3 + 1 = 4$.

In the **second sample case** $4\,2\,|\,6\,|\,1\,3\,|\,5$ is an optimal way to distribute the flowers into 4 bouquets. The total *beautiness* is $2 + 6 + 1 + 5 = 14$.